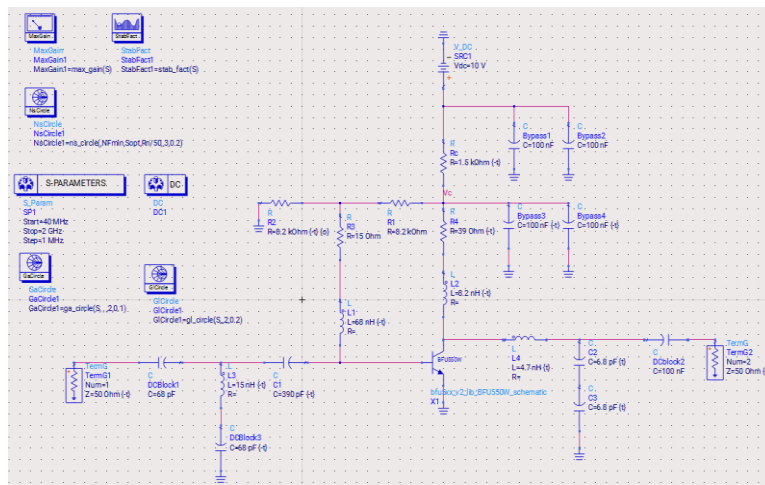


Bilkent University Fall 2025

EEE412 Microwave Electronics-Term Project Part 2 Report

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This Project focuses on design the assigned amplifier using a single BFU520W (a BJT). Choosing the operating point according to gain or noise considerations then designing stabilization network(s), input and output matching networks considering the assigned center frequency (0.68Ghz in my case) of the proposed circuit. Below schematic is the overall amplifier design on ADS with the proper input and output matching networks.



After the feedback, I needed to lower my current until about 3.5 mA (due to noise) and my Vce voltage increased to 4.12V (See Figure 3). After proper biasing I continue with RF chock design, after that ended with input and output matching and finalized the design as in Figure 2.

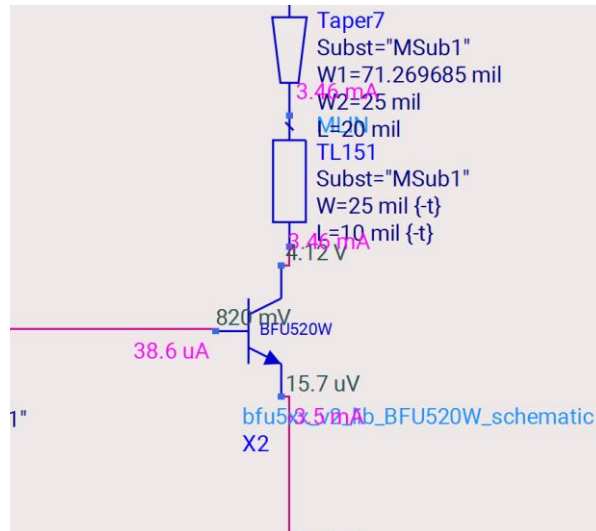


Figure 3: Biasing, DC operating Points

The amplifier has to be unconditionally stable at all frequencies (40MHz-2GHz), therefore we should check K (stability factor). Figure 4 shows the amplifier is unconditionally stable at all those frequencies.

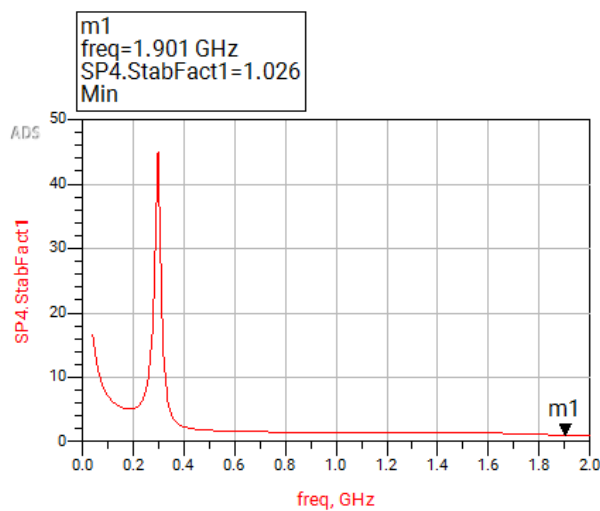


Figure 4: $K > 1$ for all Frequencies

- Center Frequency assigned = **680MHz**
- Bandwidth: 10% of the center frequency
 - **Lower limit: 646 MHz, Upper limit: 714 MHz**
- Gain > $16 - 6.5 \times f \text{ (GHz) dB}$ (f is the center frequency)
 - **Gain > $16 - 6.5 \times 0.68 = 11.58 \text{ dB}$**
- Noise Figure < $0.6 + 0.33 \times f \text{ (GHz) dB}$ (f is the center frequency)
 - **NF < $0.6 + 0.33 \times 0.68 = 0.8244 \text{ dB}$**

While I were designing I simultaneously check MAG and Noise Figure, The available gain has to be little higher than our expected gain so that, the real transferred value will be enough, if MAG is already lower than our expected value we cannot reach a higher value whatever we do.

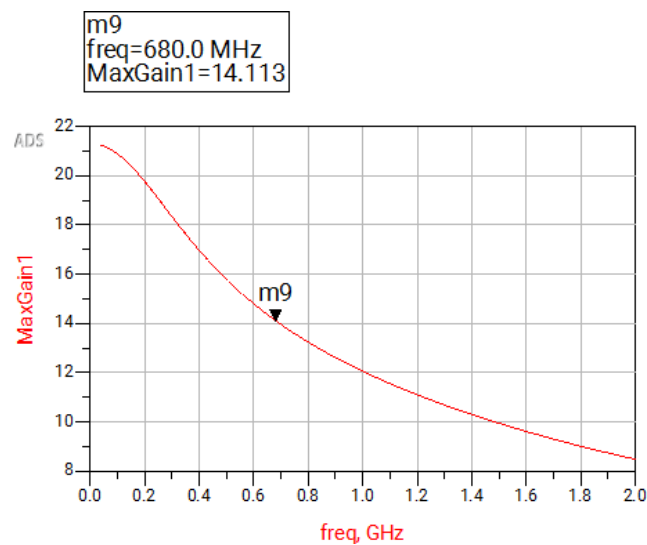


Figure 5: Maximum Available Gain Observed $\approx 14\text{dB}$ in the center frequency

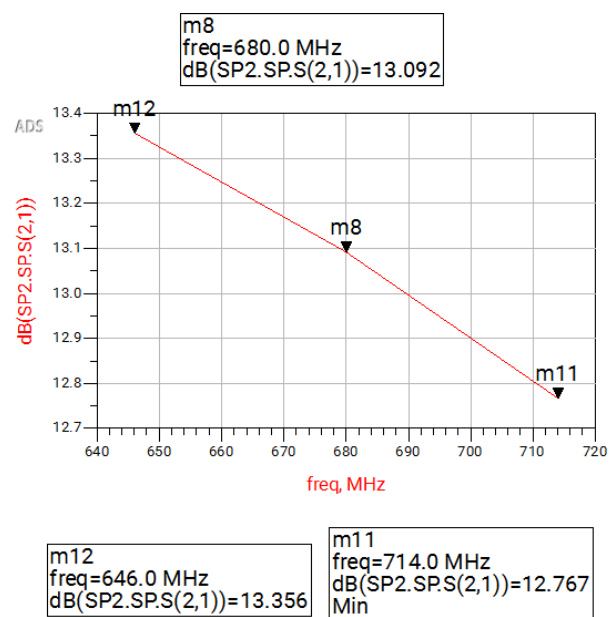


Figure 6: S21 value

At the center frequency we get **13.092dB** gain which is highly above the required gain. At the lower limit of the bandwidth we get **12.767dB** and at the higher limit we get **13.356dB**. So, we have a small gain variation <0.5dB which satisfies the specification.

Input return loss is highly above 8dB in the band (approximately **17 dB**) (See Figure 7) The worst point in the band is 646MHz with **16.874dB**. In all frequencies the worst point is close to lower limit, 646MHz. Input return loss is better at higher limit, **17.394dB**. Output return loss is approximately **12dB** which is again, satisfying the 12dB limit. See the worse point is at upper limit of the band (Figure 7). At the lower limit of the bandwidth we get **-12.385dB** and at the higher limit we get **-12.032dB**.

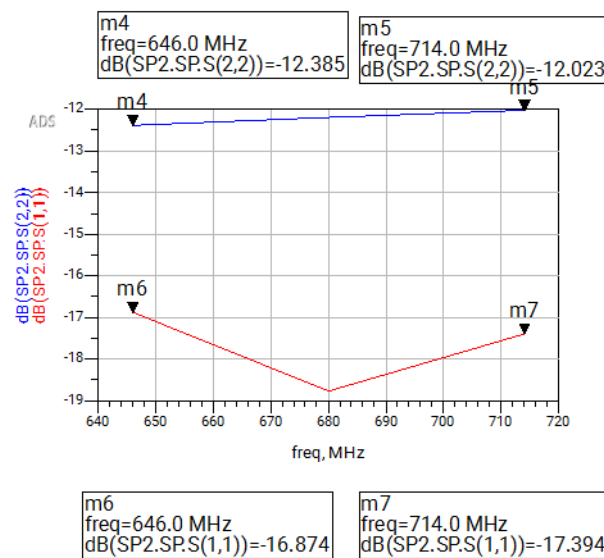


Figure 7: S22 and S11 Plot

For input output matching circuits Smith Chart helped. See Figure 8, my center frequency is close the the center which ensures the good matching.

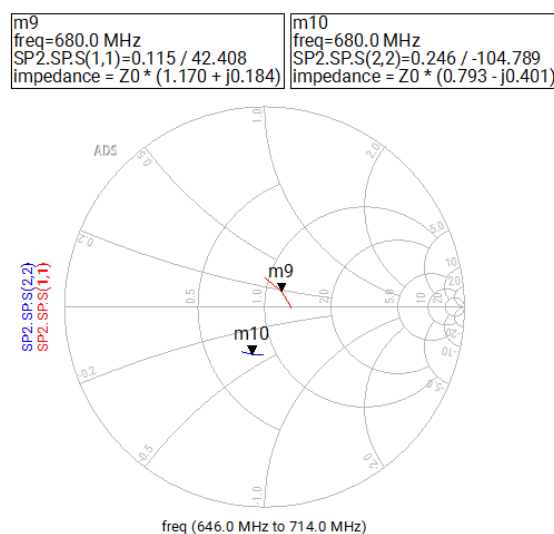


Figure 8: S22 and S11 on Smith Chart

The noise figure has to be little lower than our expected noise so that, the real noise figure will be lower enough, if NF is already higher than our expected value we cannot reach a lower value whatever we are desired to get. See below Figure gives NFmin about 0.78dB. Our NF has to be lower than **0.8244 dB**. See the worst point (**0.799 dB**) in the band is also below the range and at exactly on the center frequency we get **0.791dB**.

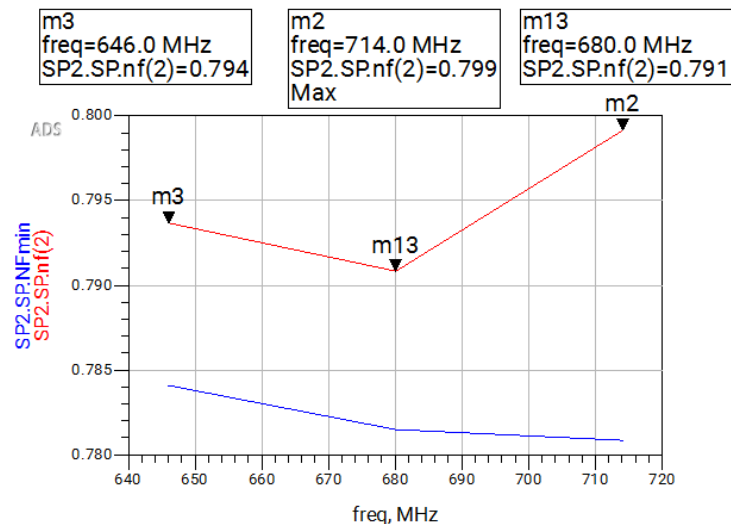


Figure 9: Noise Figure Plot in the Band

After careful trial and error small adjustments are made in order to fit the components and Mlines inside the PCB border. See below figures (Figure 10, layout), (Figure 11, top view), (Figure 12, drill view).



Figure 10: Layout, Inside the PCB Border

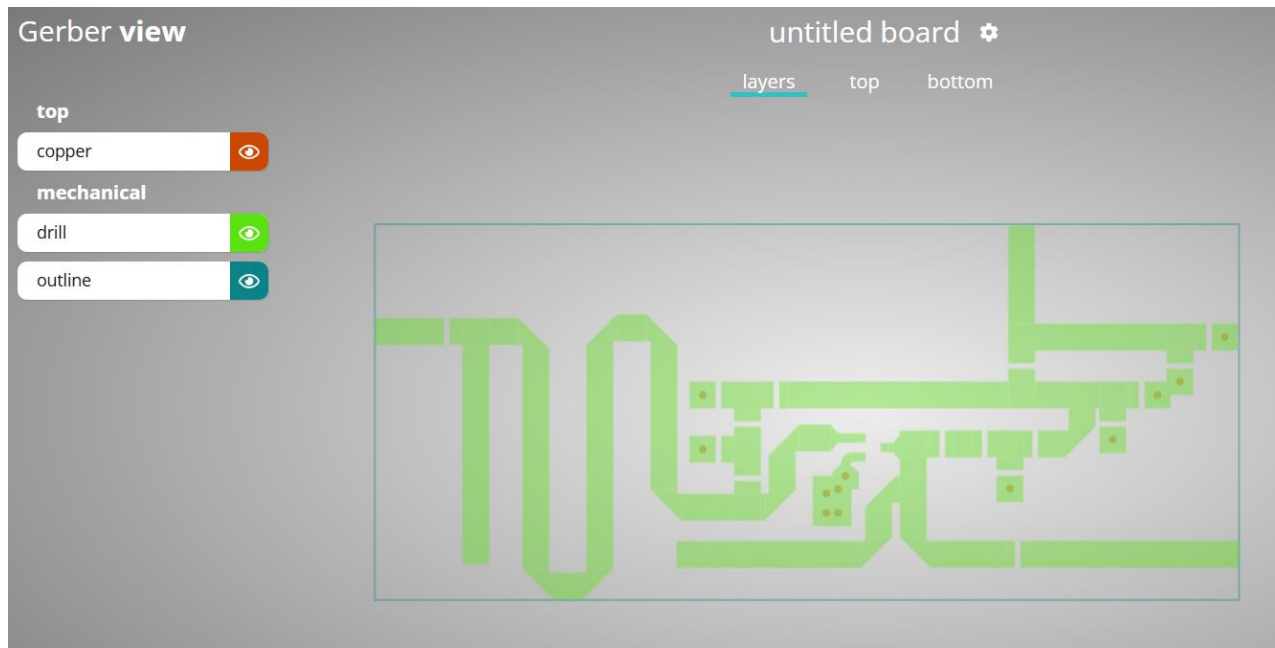


Figure 11: Top Layer of the Layout

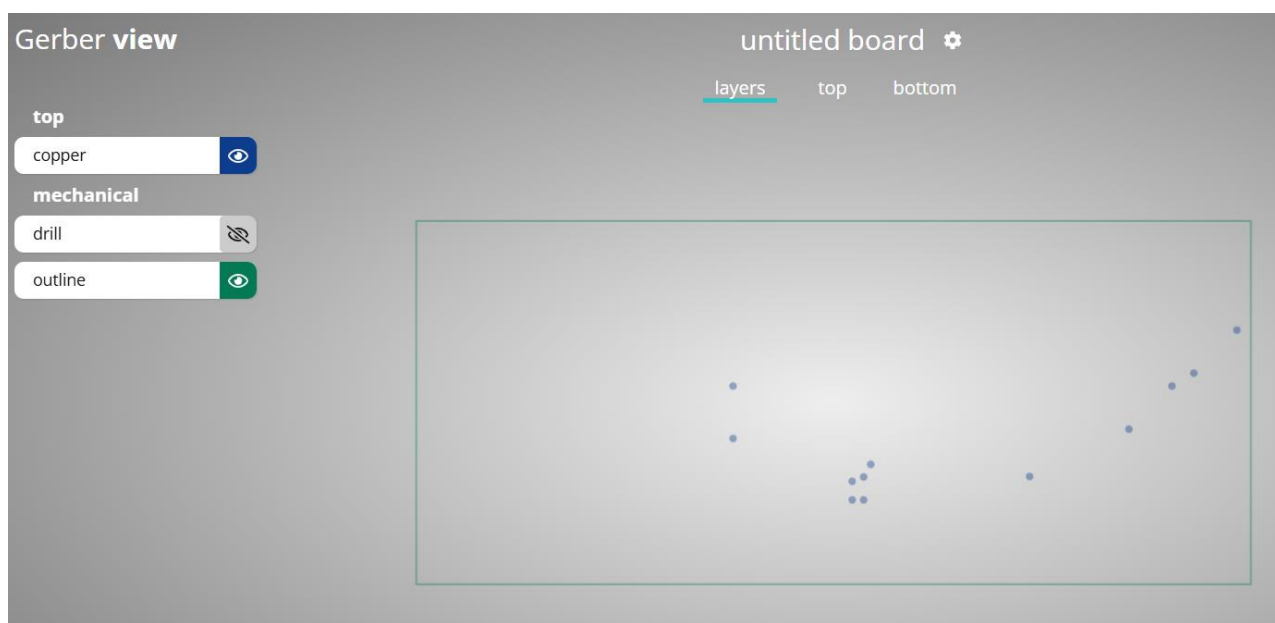


Figure 12: Drill View

[illegible]

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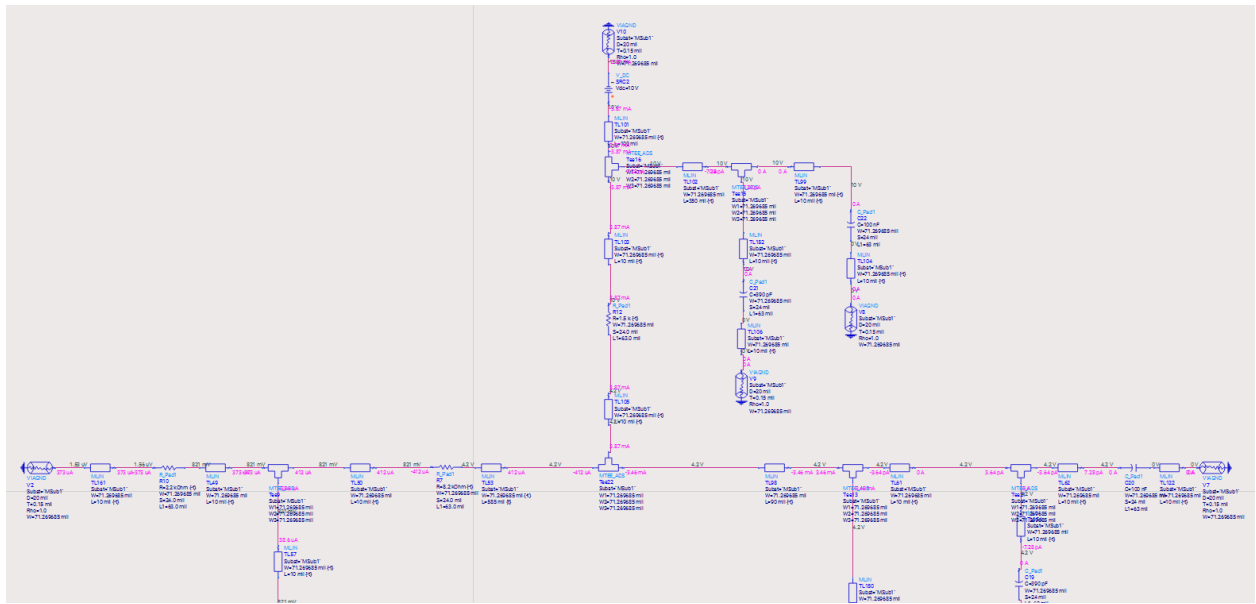


Figure 15: Biasing Network

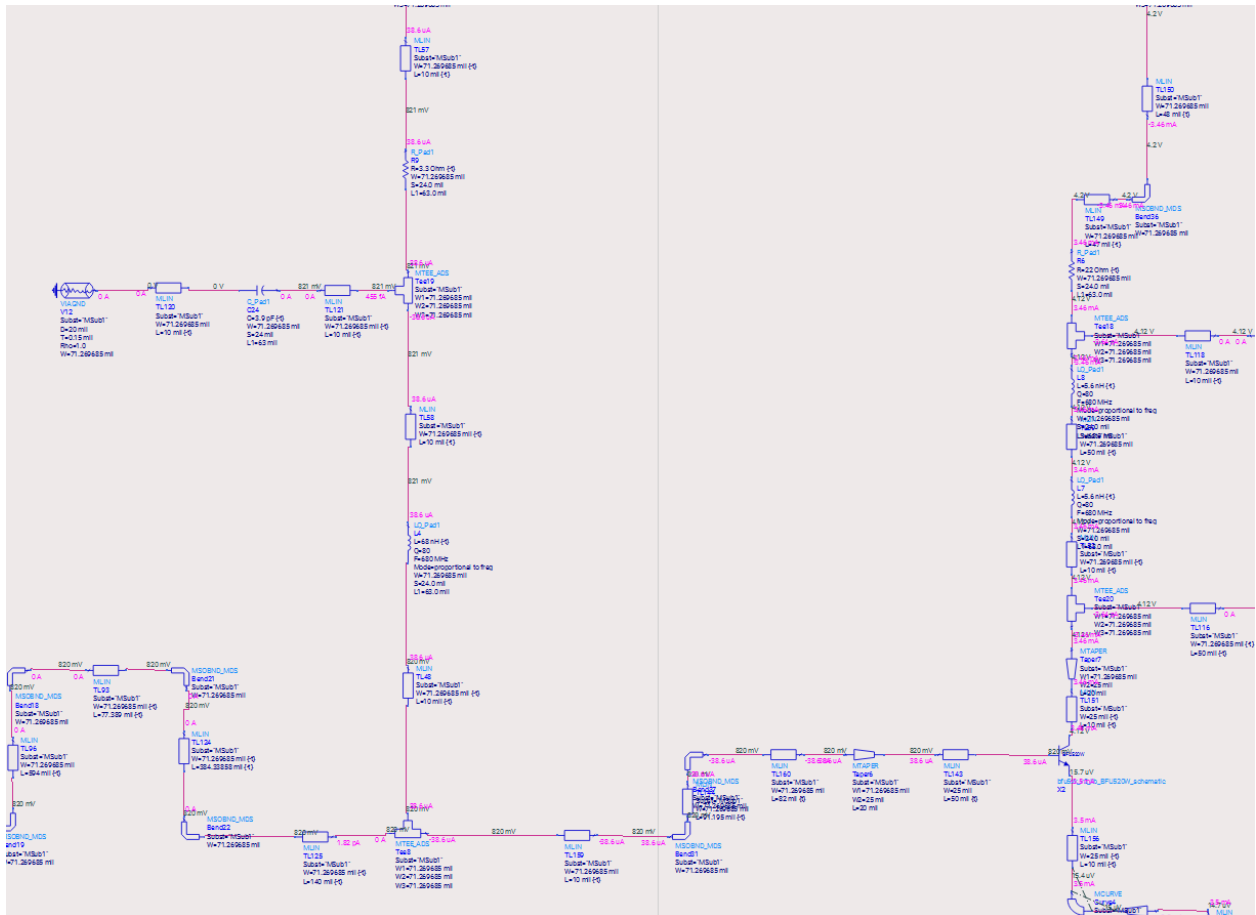


Figure 15: RF Chock